Ex.No.4	Code Completion Using a Pre-trained Transformer Model	URK23CS8018
10-09-2026		

Aim:

To implement code completion using a pre-trained Transformer model (GPT-2) and generate the continuation of partially written Python code. The experiment uses the Hugging Face transformers library and provides a Gradio-based interface for entering incomplete Python code and displaying the AI-generated completion.

Procedure:

1. Install the required libraries such as transformers, torch, gradio, pandas, and matplotlib.
2. Import the required modules from PyTorch, Transformers, and Gradio.
3. Load the pre-trained GPT-2 model using AutoTokenizer and AutoModelForCausalLM.
4. Initialize the GPT-2 tokenizer to convert the input Python code into tokens.
5. Set the model to evaluation mode using model.eval().
6. Define a code-completion function that accepts partially written Python code as input.
7. Generate the code continuation using model.generate() with suitable parameters such as max_new_tokens, temperature, and top_p.
8. Decode the generated tokens back into readable Python code and display the completion.
9. Create a Gradio interface that allows the user to enter partial Python code and view the AI-generated completion.
10. Launch and test the application by entering code such as def calculate_average(numbers): and observe the generated code continuation

Question:

Implement code completion using a pre-trained Transformer model (GPT-2) to generate the continuation of partially written Python code and provide a Gradio-based interface for code completion.

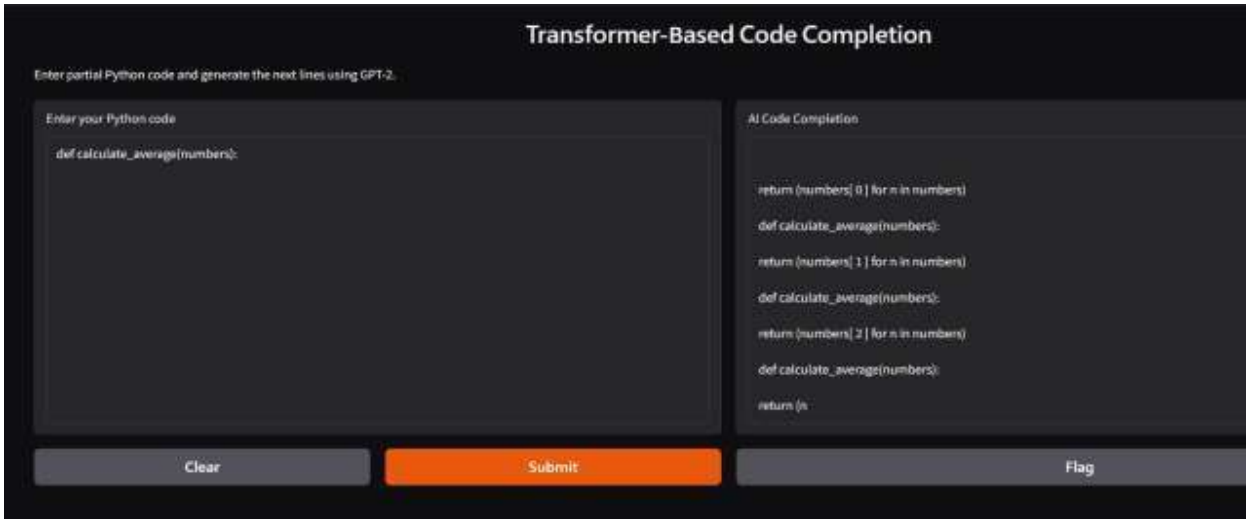
```
import torch
from transformers import AutoTokenizer, AutoModelForCausalLM
import gradio as gr
print("URK23CS8003")
model_name = "gpt2"
print("Loading GPT-2 model...")
tokenizer = AutoTokenizer.from_pretrained(model_name)
```

```
model = AutoModelForCausalLM.from_pretrained(model_name)
model.eval()
print("GPT-2 loaded successfully!")
def get_completion_only(prompt, max_new_tokens=50):
    inputs = tokenizer(
        prompt,
        return_tensors="pt"
    )
    with torch.no_grad():
        outputs = model.generate(
            **inputs,
            max_new_tokens=max_new_tokens,
            do_sample=True,
            temperature=0.2,
            top_p=0.95,
            pad_token_id=tokenizer.eos_token_id
        )
    generated_tokens = outputs[0][inputs["input_ids"].shape[1]:]
    completion = tokenizer.decode(
        generated_tokens,
        skip_special_tokens=True
    )
    return completion
prompt = """def calculate_average(numbers):
"""
print("\nGenerated code:")
print(get_completion_only(prompt))
def code_completion_interface(code):
    if not code.strip():
        return "Please enter some Python code."
    return get_completion_only(
        code,
        max_new_tokens=80
    )
```

```
demo = gr.Interface(
    fn=code_completion_interface,
    inputs=gr.Textbox(
        lines=15,
        label="Enter your Python code"
    ),
    outputs=gr.Textbox( lines=1
        5,
        label="AI Code Completion"
    ),
    title="Transformer-Based Code Completion",
    description="Enter partial Python code and generate the next lines using GPT-2."
)
demo.launch()
```

Output Screenshot:





Link

https://drive.google.com/file/d/1-51bMttQVk_SK_n8bCerYzfa_wZHj-Ex/view?usp=sharing

Result:

The pre-trained GPT-2 Transformer model successfully generated code completions for partially written Python code. The model was loaded successfully, and the Gradio interface was launched successfully.